

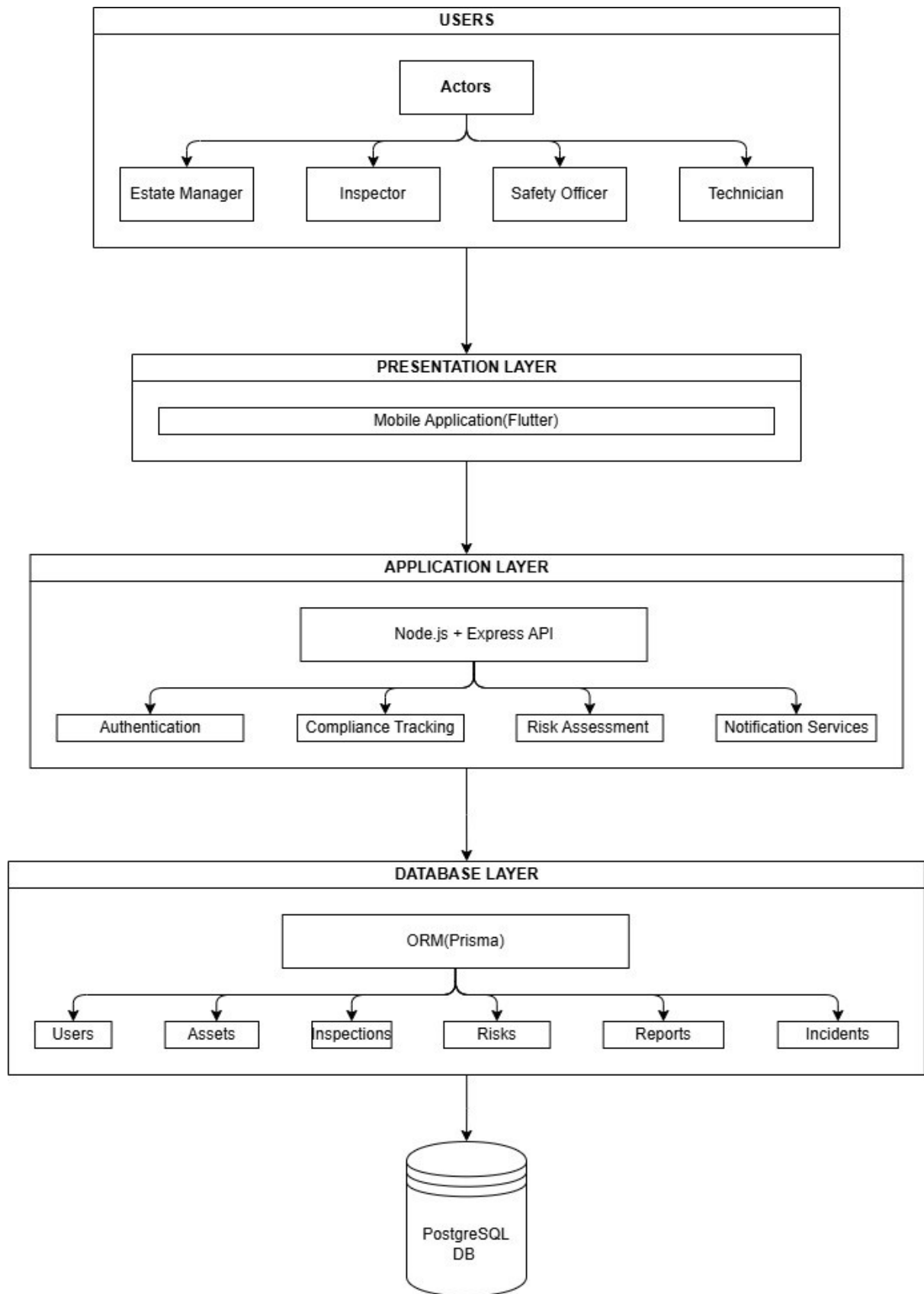
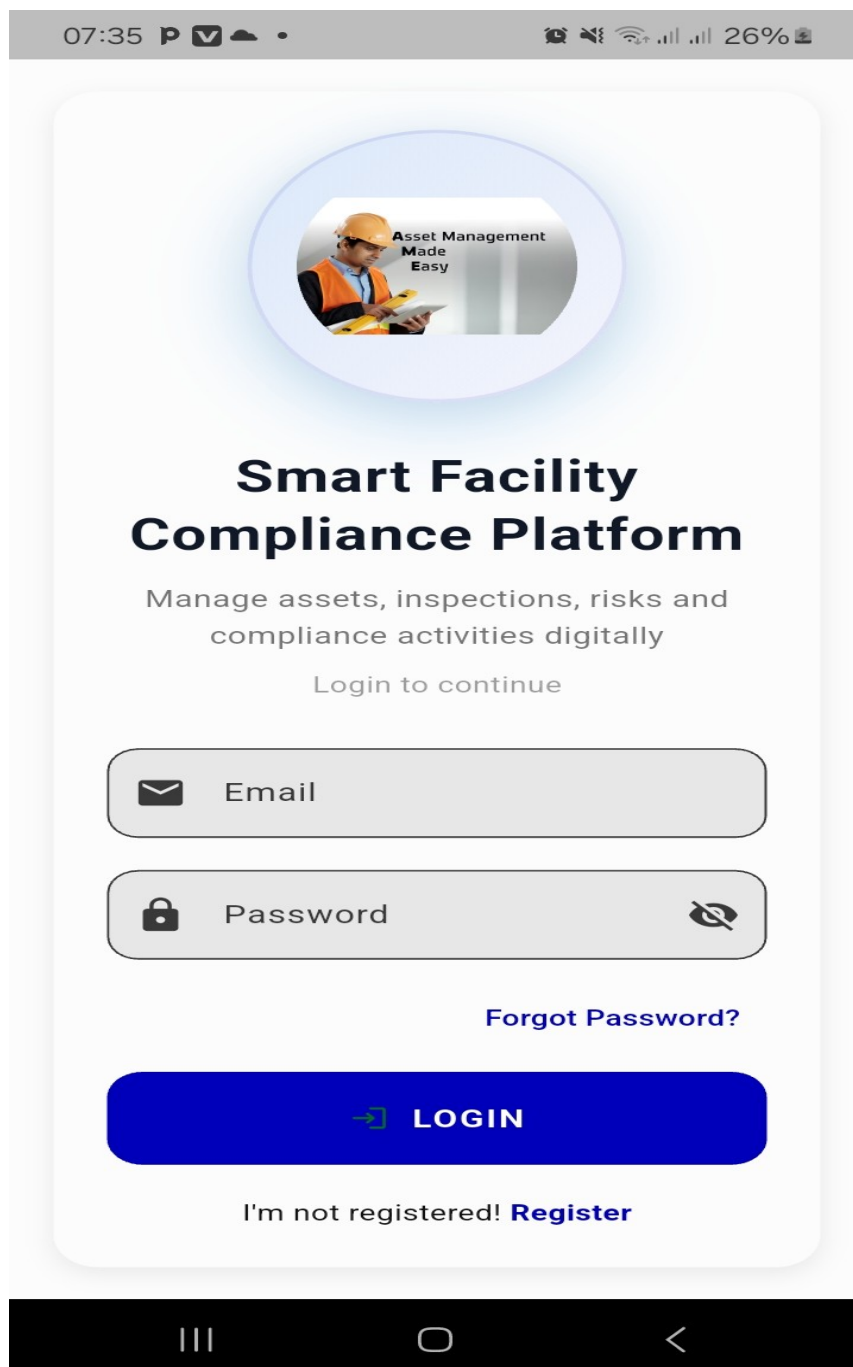


Figure 4.1 System Architecture

5.2.1 User Authentication Interface

The User Authentication Module provides secure access to the system by allowing authorized users to log in using their registered email and password. After successful authentication, the system verifies the user's role and grants access to the appropriate dashboard based on their responsibilities, such as Administrator, Inspector, Safety Officer, Technician, and Estate Officer. This ensures that users can only access the functions assigned to their respective roles.



The image shows a mobile application login screen for the 'Smart Facility Compliance Platform'. At the top, a status bar displays the time 07:35, signal strength, and 26% battery. The app interface features a header with a circular logo containing a worker and the text 'Asset Management Made Easy'. Below this, the title 'Smart Facility Compliance Platform' is prominently displayed, followed by the tagline 'Manage assets, inspections, risks and compliance activities digitally'. A 'Login to continue' prompt leads to input fields for 'Email' and 'Password'. A 'Forgot Password?' link is positioned below the password field. A large blue 'LOGIN' button with a green arrow icon is centered below the inputs. At the bottom, a link for 'I'm not registered! Register' is provided. The screen is framed by a black Android navigation bar at the very bottom.

Figure 5.1 Login Page

5.2.2 Dashboard Interface

The Dashboard Module provides users with an overview of system activities through graphical summaries and statistical information. It displays key information such as the total number of assets, scheduled inspections, compliance status, pending CAPA tasks, reported incidents, and dates nearing expiry. The dashboard enables users to quickly monitor the overall safety and compliance performance of facility assets.

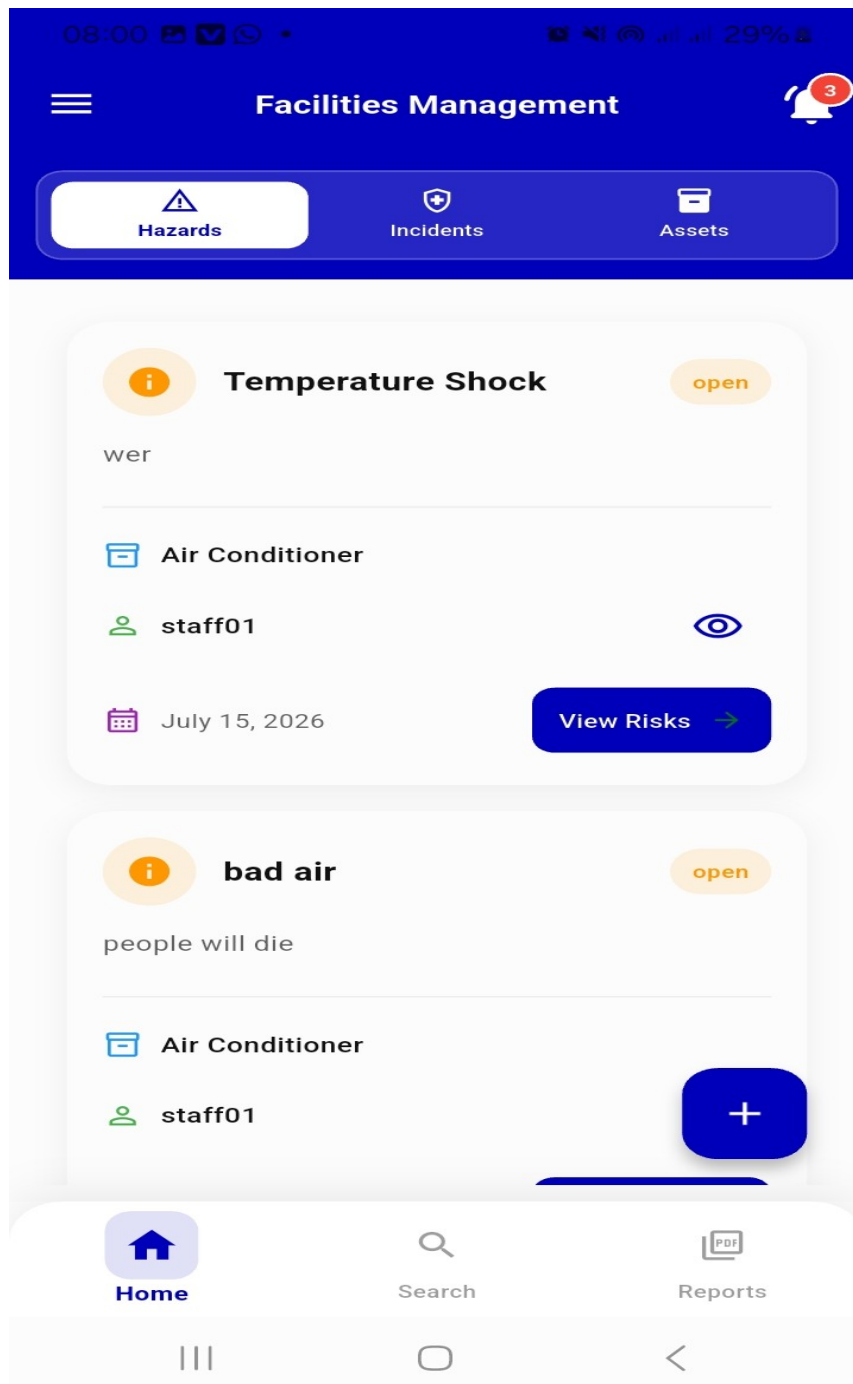


Figure 5.2: Dashboard Interface.

5.2.3 Asset Management Interface

The Asset Management Module enables users to register, update, view, and manage facility assets such as fire protection systems, lifts, electrical installations, boilers, pressure vessels, and environmental permits. Each asset is assigned unique information including its location, category, serial number, and operational status, allowing efficient monitoring throughout its lifecycle.

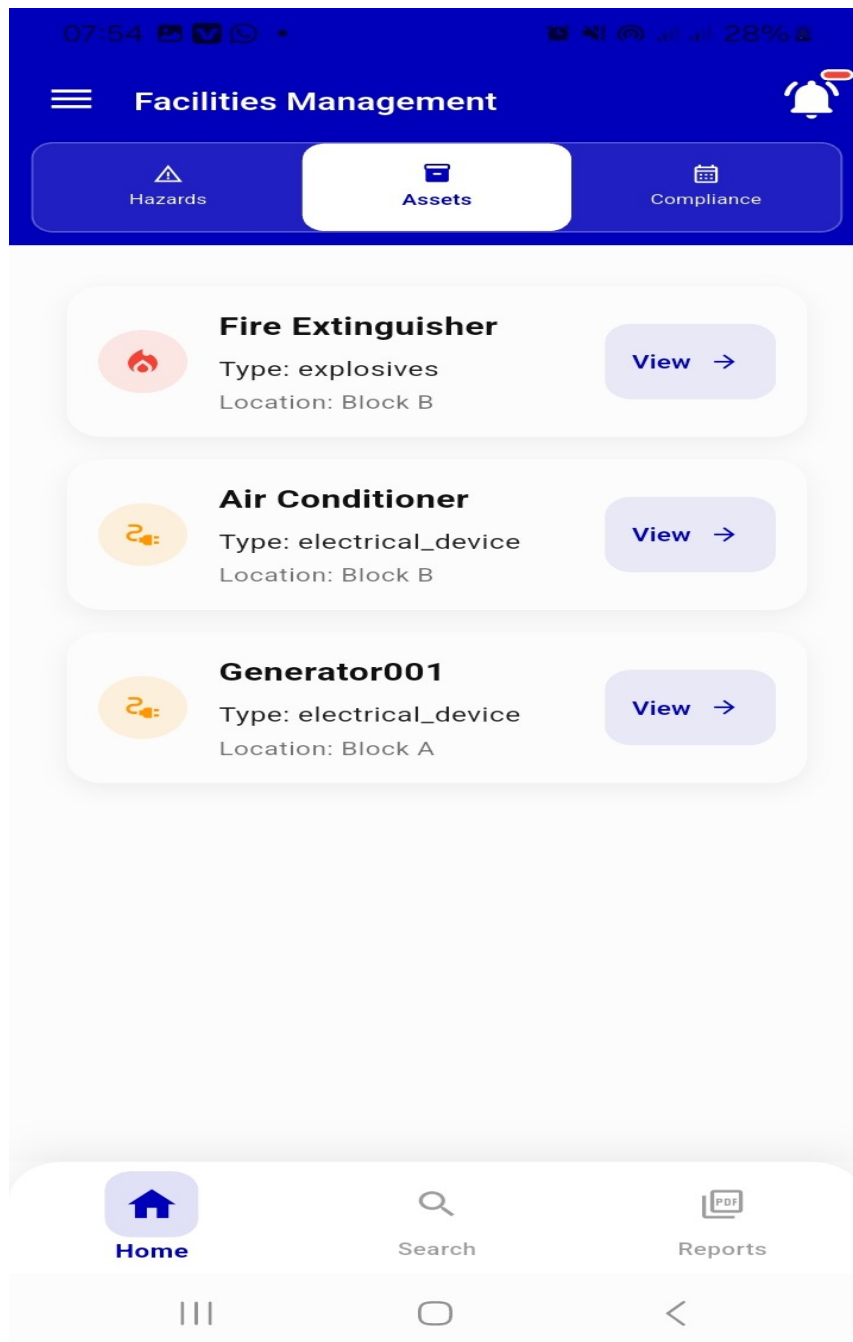


Figure 5.3 Asset Management Interface

5.2.4 Inspection Management Interface

The Inspection Management Module allows inspectors to schedule inspections, record inspection findings, upload inspection reports, and update the compliance status of facility assets. The module maintains inspection history and automatically identifies overdue inspections to support statutory compliance.



Figure 5.4 Inspection Management Interface

5.2.5 Compliance Tracking Interface

The Compliance Tracking Module monitors the compliance status of all registered assets by evaluating inspection results, certification validity, and regulatory requirements. Assets are automatically categorized as compliant, non-compliant, pending inspection, or overdue, enabling timely corrective actions.

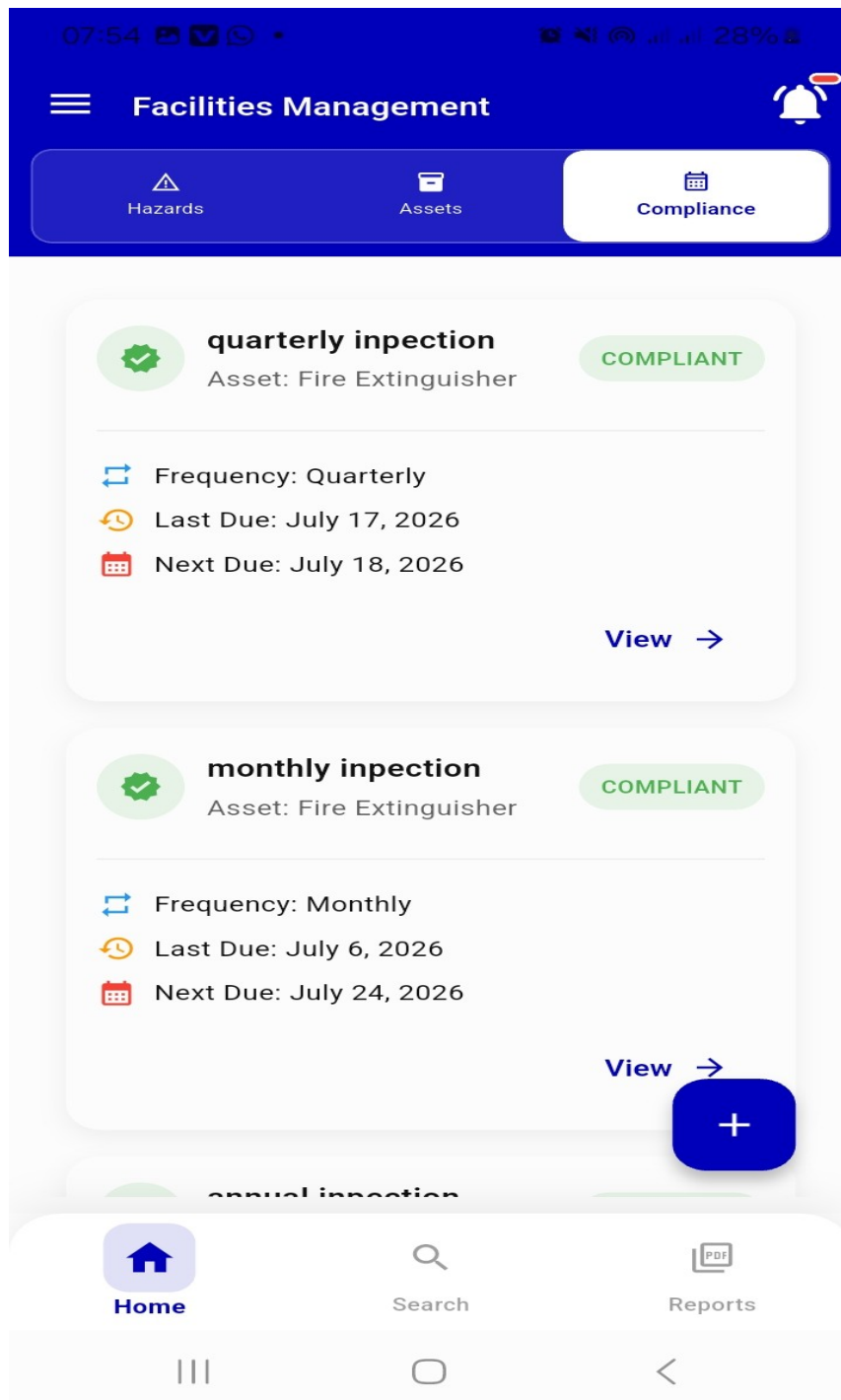


Figure 5.5: Compliance Tracking Interface.

5.2.6 Risk Assessment Interface

The Risk Assessment Module enables Safety Officers to identify hazards, assess risk levels, assign mitigation measures, and maintain a comprehensive risk register. Risks are categorized according to their likelihood and severity to assist in prioritizing safety interventions.

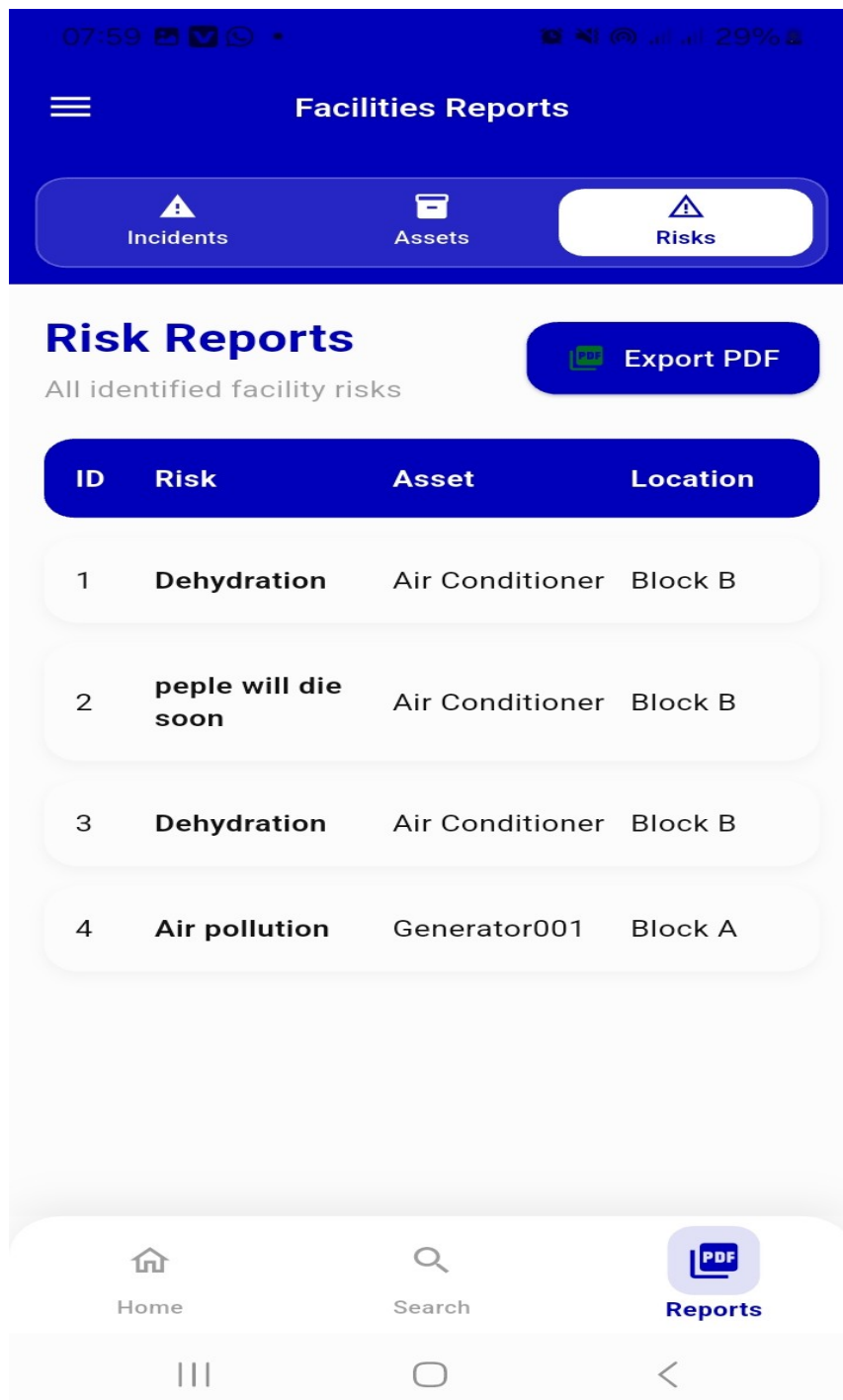


Figure 5.6: Risk Assessment Interface

5.2.7 Incident Management Interface

The Incident Management Module allows users to report incidents and near-miss events associated with facility assets. The system records incident details, affected assets, severity level, supporting evidence, and investigation results, facilitating effective incident management and future analysis.

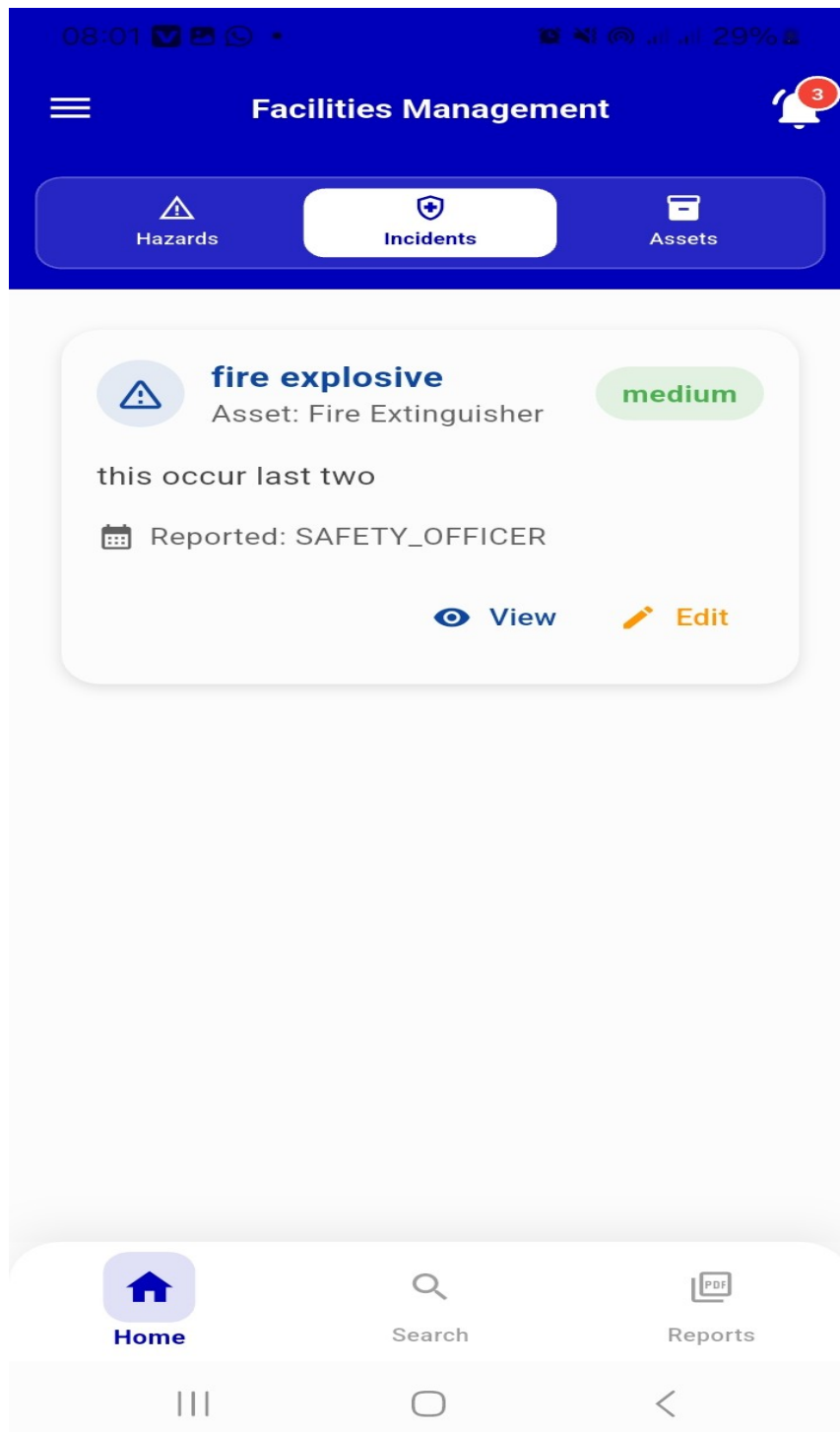


Figure 5.7: Incident Management Interface.

5.2.8 CAPA Management Interface

The Corrective and Preventive Action (CAPA) Management Module enables Safety Officers to create corrective and preventive action tasks after risks, incidents, or inspection findings are identified. CAPA tasks are assigned to technicians, monitored until completion, and verified before the asset's compliance status is updated.

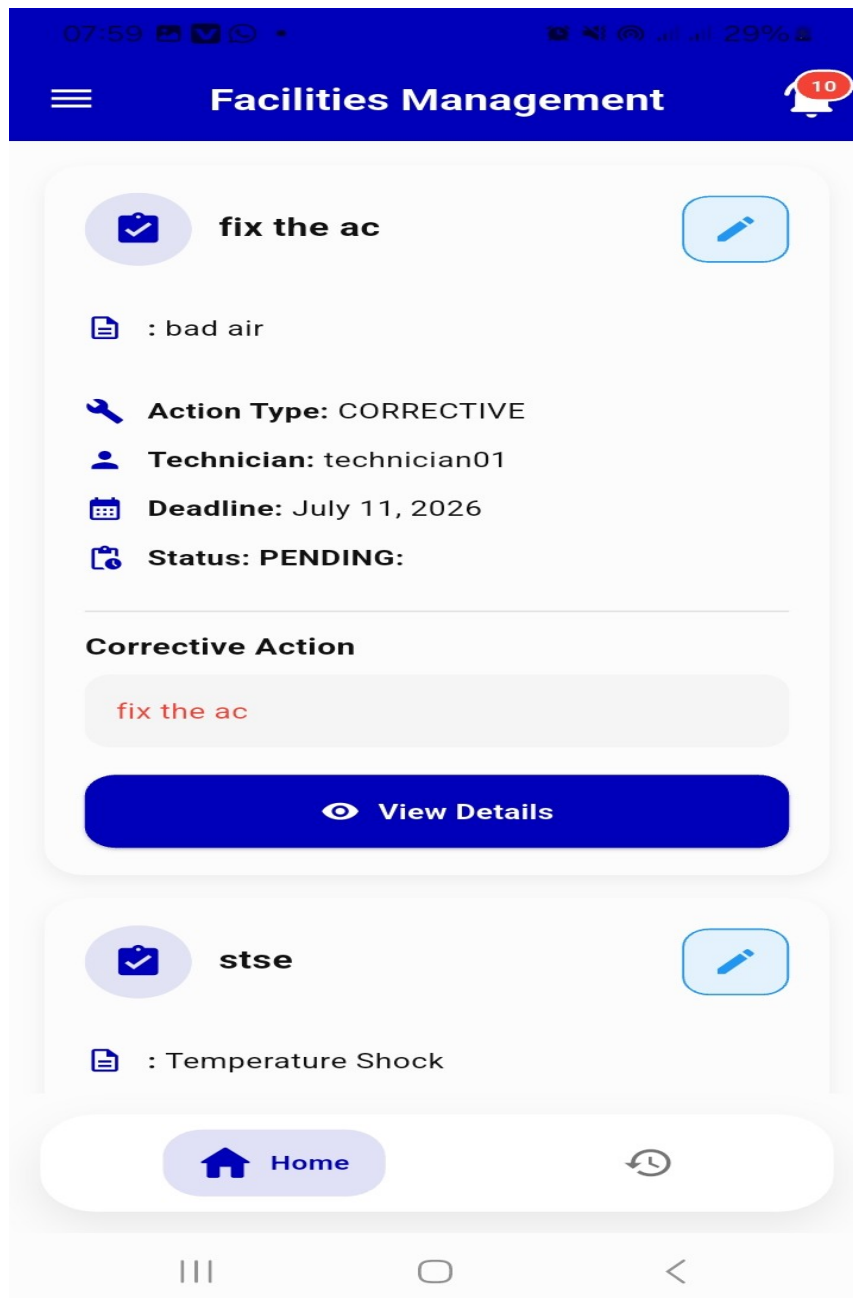


Figure 5.8: CAPA Management Interface.

5.2.9 Notification Interface

The Notification Module automatically informs users about important events, including upcoming inspections, overdue CAPA tasks, expiring certificates, and newly assigned responsibilities. Notifications are delivered through the system interface to improve communication and ensure timely completion of compliance activities.

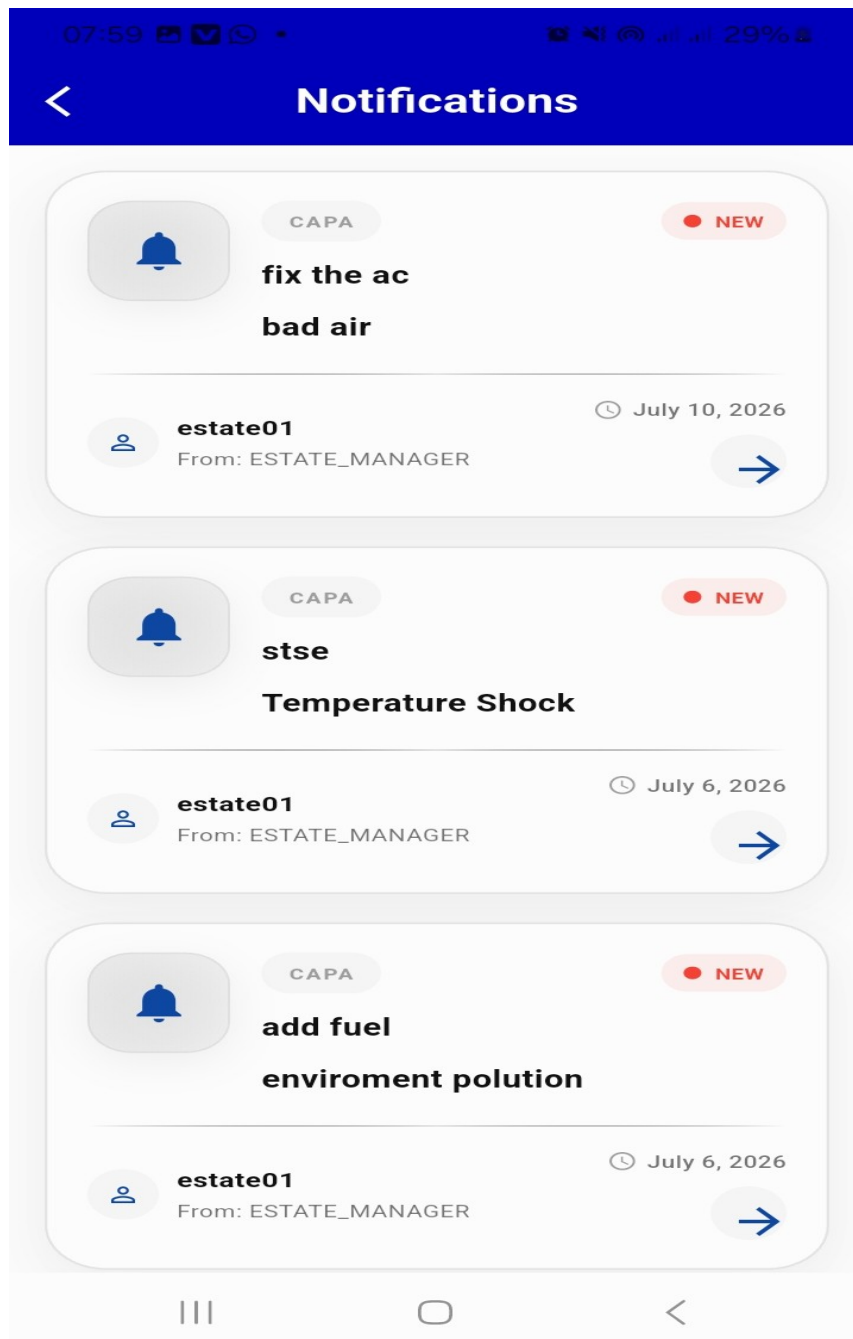


Figure 5.9: Notification Interface.

5.2.10 Reports Interface

The Reports Module generates comprehensive reports on inspections, compliance status, incidents, risk assessments, CAPA activities, and certification records. These reports assist management in monitoring system performance, supporting audits, and making informed decisions regarding facility safety.

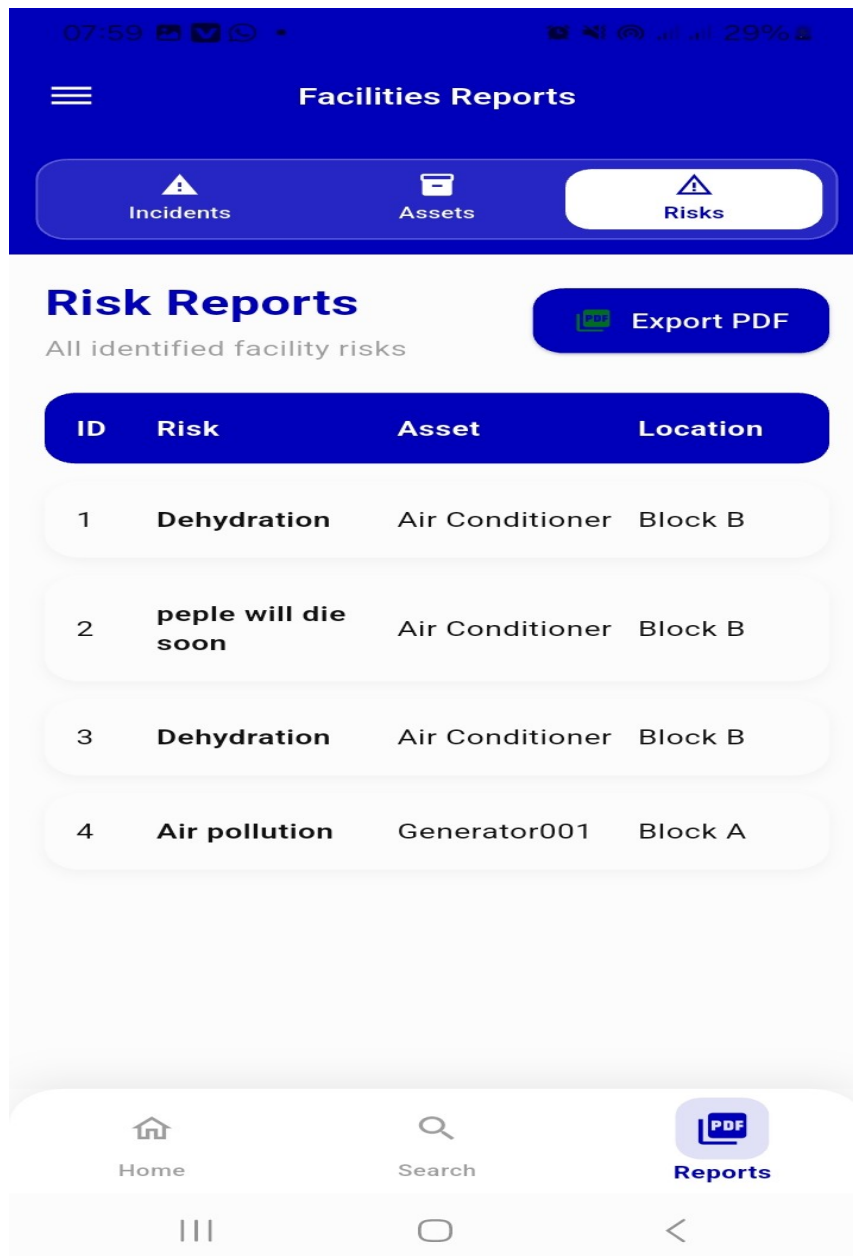


Figure 5.10: Reports Interface.